Deep Learning Lab : Capstone Project Proposal

Project Title : Multimodal Health State Prediction Using Smartphone Sensors

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1) Motivation

- Growing health issues require early monitoring and prevention
- Traditional medical tests are costly and not continuous
- Smartphones provide accessible health-related sensor data
- Deep learning enables accurate pattern recognition
- Need for affordable real-time health prediction systems

2) Problem Statement

Existing health monitoring applications rely on single data sources, resulting in inaccurate and delayed detection of early health conditions. Issues such as dehydration, fatigue, stress, and anemia often remain unnoticed due to lack of continuous monitoring. There is a need for an intelligent multimodal system that integrates smartphone sensor data using deep learning for early health risk prediction.

3) Objectives

- Develop a multimodal deep learning health prediction system
- Integrate camera, audio, and motion sensor data
- Detect early health risks accurately
- Improve prediction performance using sensor fusion
- Enable real-time mobile-based monitoring

4) Methodology

Step 1: Collect multimodal data from smartphone sensors
Step 2: Preprocess data through noise removal and normalization
Step 3: Extract features using deep learning models
Step 4: Fuse sensor features using attention-based learning
Step 5: Train and evaluate health prediction model

5) Expected Outcomes

- Early detection of dehydration, fatigue, stress, and anemia
- Improved prediction accuracy and reliability
- Real-time health alerts
- Practical preventive healthcare solution

6) Dataset Used

- Smartphone camera images for physical indicators
- Audio recordings for breathing and stress analysis
- Accelerometer data for activity monitoring
- Public biomedical datasets for validation

7) References

1) Review of Multimodal Machine Learning in Healthcare

🔗 https://www.sciencedirect.com/science/article/pii/S1566253524004688

2) Multimodal Integration in Health Care

🔗 https://www.jmir.org/2025/1/e76557

4) Real-Time Health Monitoring Using Deep Learning + IoT

🔗 https://www.jneonatalsurg.com/index.php/jns/article/view/6866

5) Classic Multimodal Deep Learning Review (Foundational)

🔗 https://www.nature.com/articles/s41746-022-00712-8